HEAPLINE · MARKET AND VIABILITY

HeapLine market and viability

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	First market note, written for the kit 1.9.1 upgrade; prices cited with the date they were checked

HeapLine market and viability

The users are waste pickers' cooperatives and the site operators, municipal units and Red Cross branches that stand behind them; nothing was found that packages a probe, board, stretcher and capstan kit for a dumpsite, so the benchmarks below price single items only. An open design lets a local workshop build and mend the kit from stock steel, timber and rope for an estimated USD 827 a site plus USD 547 for each shared capstan set, and lets it spread through cooperatives, NGOs and workshops without a licence fee. Method: STANDARDS section 23.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Waste pickers' cooperatives and associations	Active dumpsites with homes at the toe, for example Kiteezi (Kampala) and Koshe (Addis Ababa) (IDS, 2017)	Avalanche probe, one 320 cm: Black Diamond Quickdraw Pro 320, USD 74.96, retailer page checked 2026-11-03 (evo); HeapLine has six probes in the kit	Search kit USD 827 a site (HPL-CAL-001 G1)	A site-kept kit with a plan, lookout and stop rule that people on site can use in the first minutes
Site operators and municipal staff	Sites awaiting closure or rehabilitation	Folding pole stretcher: USD 179.00 (350 lb rating), retailer page checked 2026-11-03 (SOS Products); HeapLine's HDPE sheet stretcher is BOM line 23, USD 82	Host site USD 1,374 (search kit and the shared capstan set)	A low-cost emergency measure that can be stocked and inspected monthly
Red Cross branches, fire and police responders	Handover from community search to formal rescue	A 2 ton hand cable puller (come-along), USD 83.99, listed as discontinued, retailer page checked 2026-11-03 (Paul B Hardware); for scale only: it is a different, wire-rope machine, not tested against the 5 kN pull here	Capstan set USD 547 shared by neighbouring sites (HPL-CAL-001 G1b)	Arrive to a search that is organised, with marks placed
Local fabricators	Workshops near dumpsites	No kit price found for a made, site-held slide-rescue kit: to assess	Parts only; labour and workshop margin to assess	A paid build and repair service from open drawings

Open route

Cooperatives and NGOs would borrow or buy a kit, local fabricators would build it from `bom/bom.csv` and the drawings, and universities and makerspaces could build a test unit. The hardware is under CERN-OHL-S-2.0 and the calculators under MIT (`REUSE.toml`), so any maker can share improvements back under the same terms. Whether a commercial supplier would ever sell kits is Amish's decision and is not assumed here.

Viability

On paper the parts cost is USD 426 under the USD 1,800 value-engineering target (a hypothetical control target, not a limit). Demand is episodic and tied to rainy seasons and to cooperatives' ability to inspect the kit monthly. Unknown and all "to assess" with a co-design partner: willingness to pay, how many sites would share a capstan, local labour rates and who pays for the monthly inspection. No price in this note is a quote; unit costs come from `bom/bom.csv`, which is indicative.

Sources

- IDS, Recognise informal waste pickers to avoid future disasters like Koshe, 2017, https://www.ids.ac.uk/opinions/recognise-informal-waste-pickers-to-avoid-future-disasters-like-koshe/.
- evo, Black Diamond Quickdraw Pro 320 Probe, USD 74.96, https://www.evo.com/products/200661-black-diamond-quickdraw-pro-320-probe, checked 2026-11-03.
- SOS Products, Folding Pole Stretcher 122795, USD 179.00, https://www.sosproducts.com/Folding-Pole-Stretcher-p/122795.htm, checked 2026-11-03.
- Paul B Hardware, Maasdam Powr-Pull 2 ton cable puller MP144SB6, USD 83.99, discontinued, checked 2026-11-03 (shared with SiltHaul).